

ARAB YOUTH SPACE HACKATHON 2026 · CHALLENGE 813 · PROOF OF CONCEPT

Madar Oman

Flood evidence through cloud — with a stated noise floor.

Team T0045 | Theme 01 — Climate Adaptation and Disaster Recovery | Sultanate of Oman

THE PROBLEM

Oman's floods arrive under cloud — and radar has a trap here

2 weeks

After the flash floods of 14–15 April 2024, the first usable Sentinel-2 scene came on 29 April, with 17–75 % cloud.

Dry sand looks like water

At C-band, fine dry sand is as smooth and dark as standing water. A simple threshold over northern Oman reports floods that never happened.

No stated confidence

Flood maps are published without saying how much of the mapped water would also appear on a dry day.

Who is affected: residents of the Al Batinah coast and the wadis of Ash Sharqiyah; civil defence, municipalities and road authorities deciding where to go first.

BUSINESS USE CASE

The duty analyst needs to know: is this map an event, or noise?

End user and decision

- Operations rooms in civil defence and municipalities: which roads and neighbourhoods first.
- Engineers and loss adjusters: which repairs and claims the evidence supports.
- Today they rely on field calls, late optical images and global products not tuned for desert terrain.

What we offer

- Subscription — national operations centre and analysis seats for public bodies.
- Per-event reports — for insurers and consultancies.
- On-premise deployment — for agencies with data-residency rules.
- Core data is open; our costs are compute and people.

Madar Oman is not a warning system. Oman's official authority is the CAA National Multi-Hazard Early Warning Centre; we provide the evidence layer beneath response and recovery.

DATA

Open Earth-observation data, each for one job

DATASET	PROVIDER	USED FOR
Sentinel-1 RTC, VV, 20 m	ESA / Copernicus	Flood extent through cloud; baseline per viewing geometry
Sentinel-2 L2A, 10 m	ESA / Copernicus	Vegetation and moisture indices; shoreline year by year
Copernicus DEM GLO-30	ESA / Copernicus	Slopes over 5° excluded (radar shadow)
ESA WorldCover 10 m	ESA	Permanent water excluded; built-up exposure
ERA5	ECMWF / Copernicus	Rainfall context for drought

Hyperspectral — stated plainly

Our headline result is a radar result and uses no hyperspectral data. We restrict every spectral index in code to Satellite 813's 400–1700 nm range, so an index 813 cannot compute is refused, not borrowed from another sensor. We have located one EnMAP before/after pair over Ibra / Al Qabil (fifteen months apart: geomorphic change only). We have not processed 813 data yet.

APPROACH

Flood is water now that is not water normally — and more than an ordinary day shows

01
Read
10 Sentinel-1 VV passes, one viewing geometry, 20 m grid

02
Baseline
Median of 5 earlier passes: what the ground normally is

03
Mask
VV < −16 dB now, not in baseline, not steep, not permanent water, ≥ 0.5 ha

04
Split
Passes before landfall are ordinary days

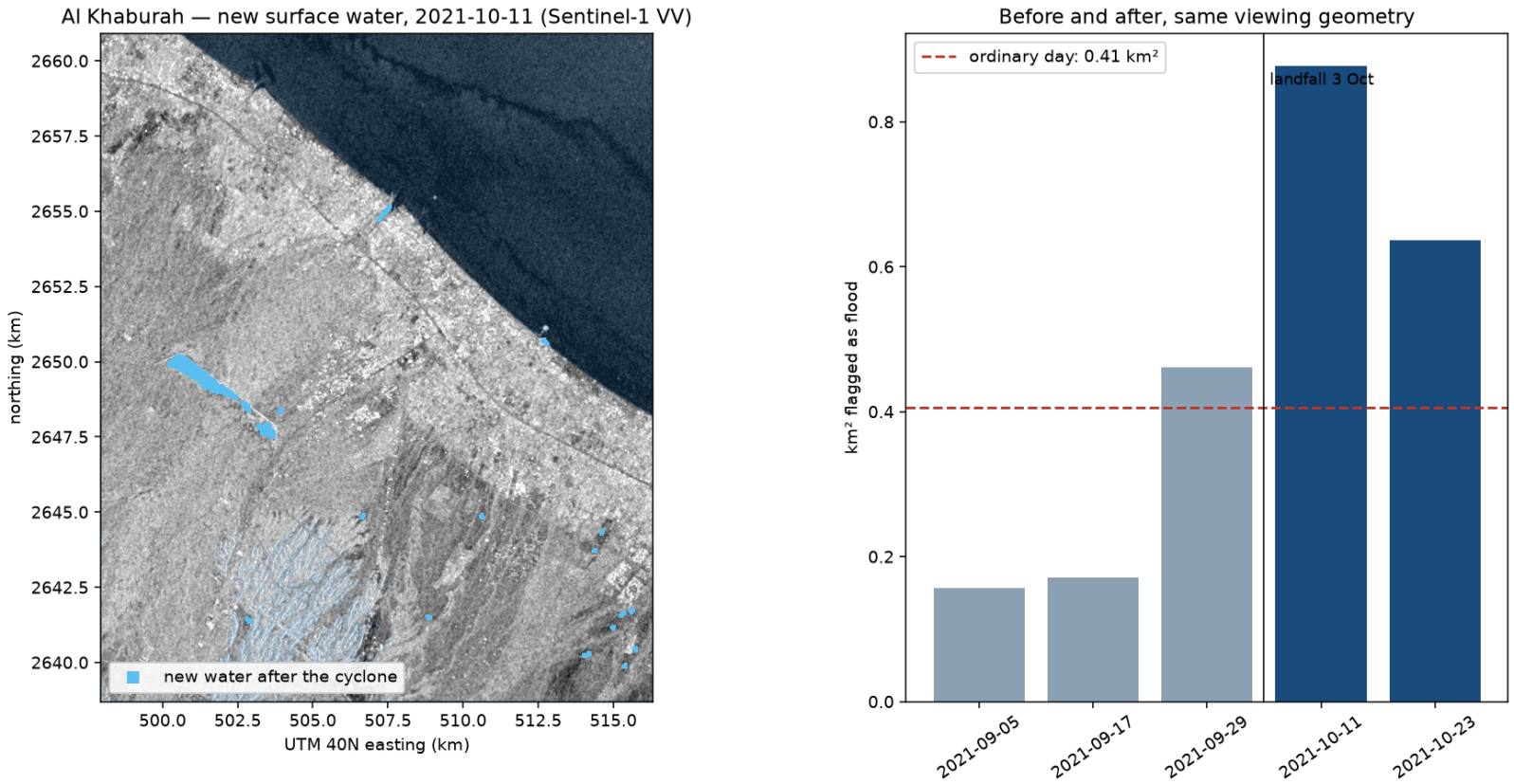
05
Subtract
Remove every cell flagged before the event → new water

06
Null test
Same subtraction on a dry day = noise floor. Declare only above it

Deterministic: no trained model, no random numbers. When there are too few passes for a baseline, or none after the event, the analysis stops and says what it needs.

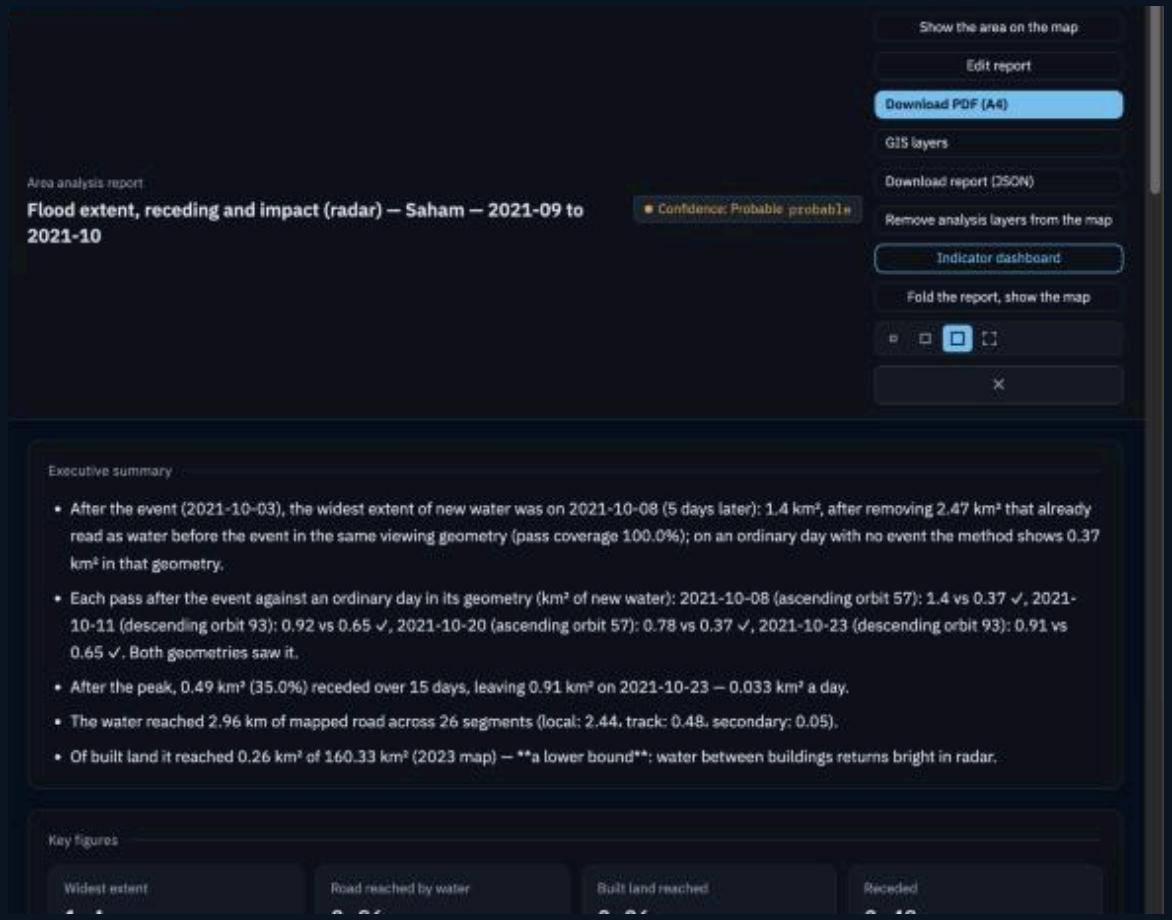
CASE 1 · FLOOD · REPRODUCED BY THE NOTEBOOK

Al Khaburah after Cyclone Shaheen: 0.88 km² of new water, against 0.41 km² on an ordinary day



Sentinel-1 VV, 11 October 2021, descending orbit 93. Left: new water (blue) over backscatter; the large patch is the Wadi Al Hawasnah dam reservoir filling. Right: area flagged per pass; dashed line = noise floor. Output of notebooks/01_flood_shaheen_khaburah.ipynb.

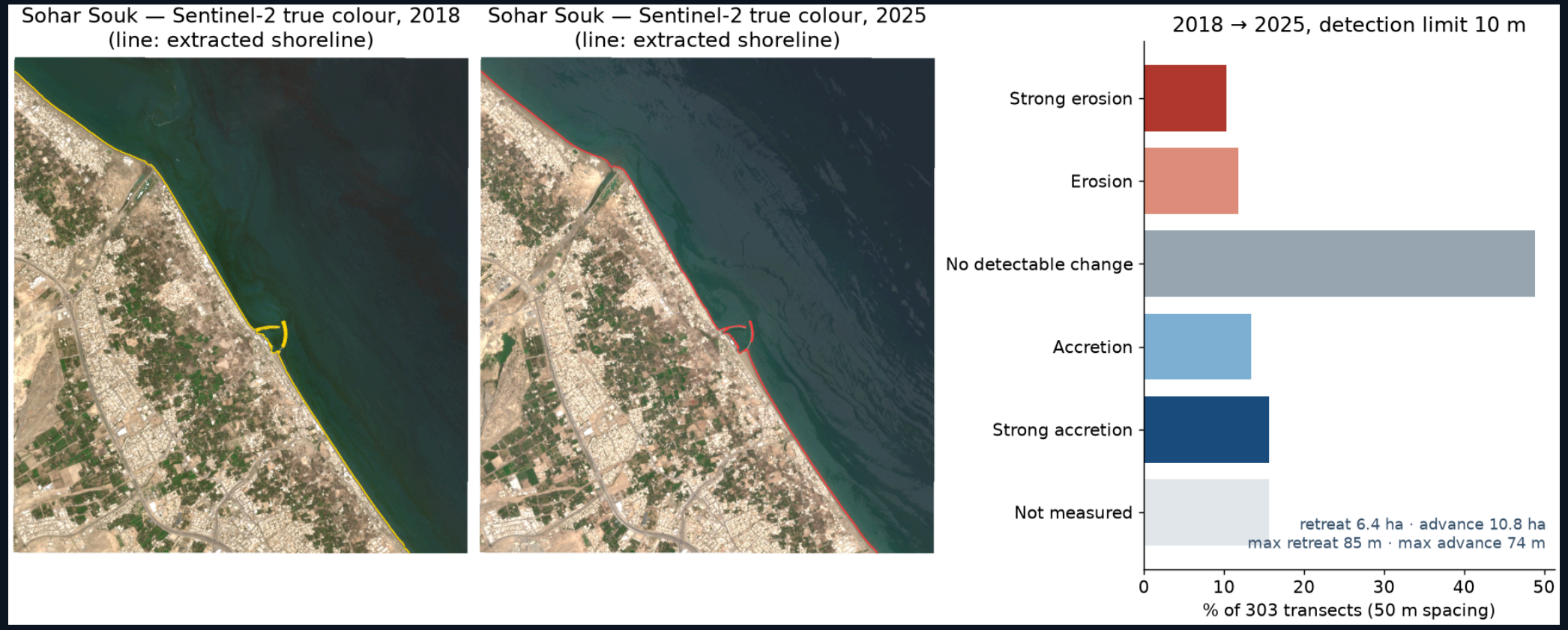
Saham: 1.40 km² of new water against 0.37 km² — with roads and built land inside it



What the report states

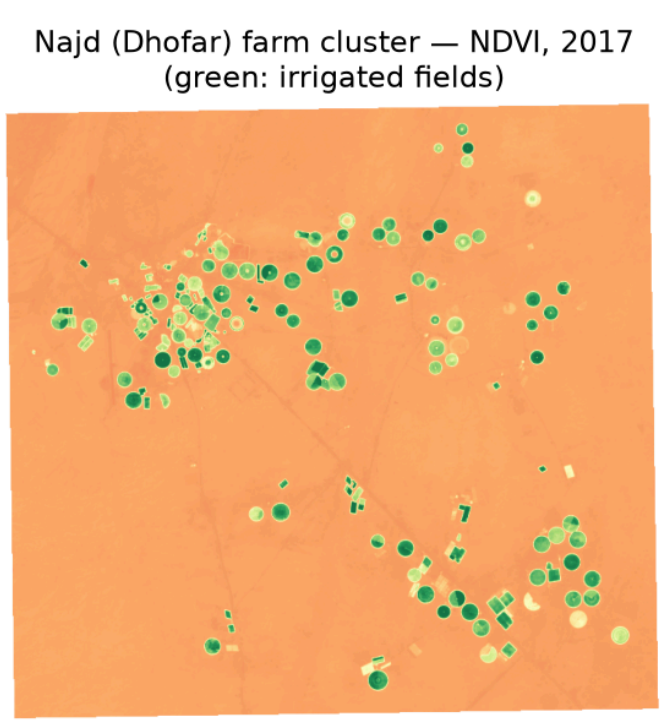
- **1.40 km²** of new water on 8 Oct 2021, five days after landfall
- **0.37 km²** on an ordinary day, same method
- **2.96 km** of mapped road reached
- **0.26 km²** of built land reached — a lower bound
- **35 %** receded within 15 days
- Screenshot of the area report in the running platform.

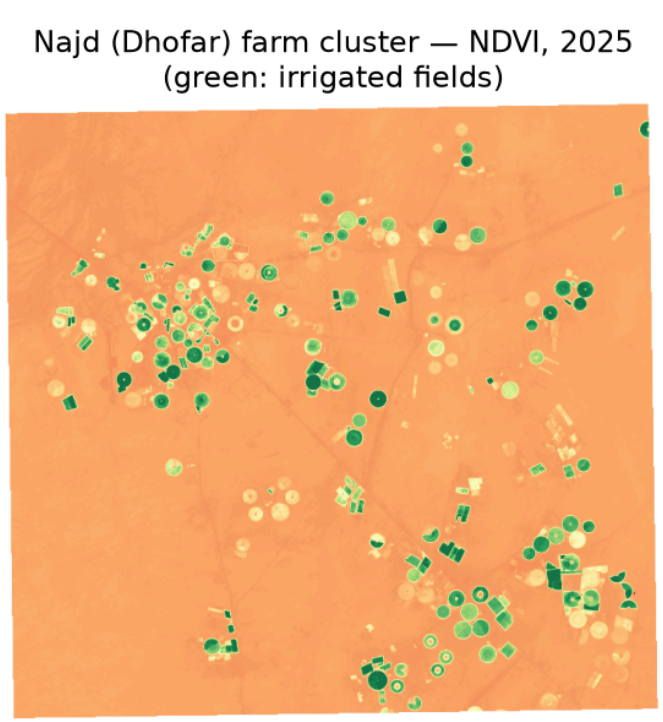
Sohar Souk shoreline, 2018 → 2025: half the coast shows no detectable change

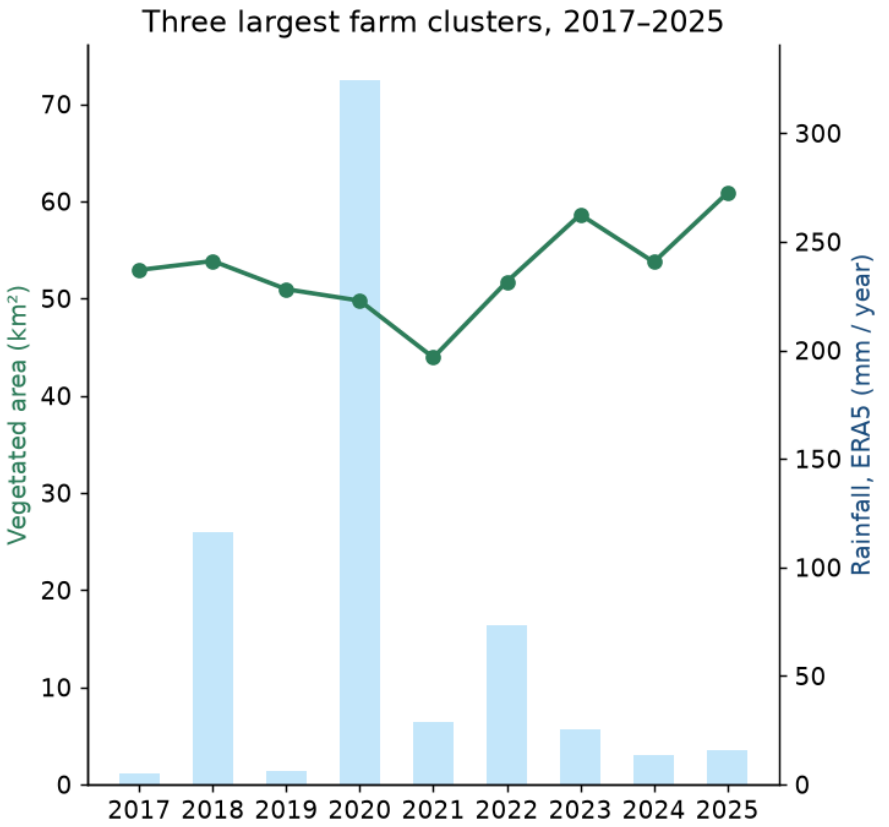


Sentinel-2. 303 transects at 50 m spacing over 15.1 km; detection limit 10 m (one pixel). 49 % no detectable change, 29 % accretion, 22 % erosion; retreat 6.4 ha, advance 10.8 ha. Compared on site with surveyed RTK points.

Najd farms, 2017 → 2025: vegetated area between 44 and 61 km², tracked against rainfall







Sentinel-2 NDVI and ERA5 rainfall for the three largest farm clusters. Lowest vegetated area 43.9 km² in 2021, highest 60.9 km² in 2025. Irrigated from groundwater: the area does not follow rainfall, which is the finding.

THE PLATFORM

One platform behind every case: draw an area, choose an indicator, get a sourced report



What is running today

- National risk table: 61 wilayats, ten hazards, every score decomposable into its factors.
- Analysis workspace: flood, drought, coastal change and spectral indices on any drawn or uploaded area.
- Reports in Arabic and English, HTML and PDF, with sources and limits.
- Thirteen data feeds; twelve need no credentials.
- Basemap shown: Sentinel-2 cloudless by EOX.

VALIDATION AND LIMITATIONS

We report "detected", "not distinguishable" and "cannot tell"

AREA, CYCLONE SHAHEEN 2021	NEW WATER	ORDINARY DAY	READING
Al Khaburah town and coastal plain	0.88 km²	0.41 km²	Detected, in both viewing geometries
Saham wilayat	1.40 km²	0.37 km²	Detected
Al Musannah wilayat (landfall)	2.17 km²	2.03 km²	**Not distinguishable from noise**
As Suwayq, Sohar	—	—	Too few usable passes; analysis refused

How we validated

The null test, and agreement between two independent viewing geometries. That is internal consistency — not ground truth.

Where it breaks

First usable pass eight days after landfall; water between buildings is invisible to radar; rough or vegetated water is missed; no field survey of this event.

IMPACT

Evidence that arrives while it still changes decisions

Societal

A defensible answer to "which roads and neighbourhoods are inside the water" in the days when cloud blocks every optical satellite.

Economic

Evidence for recovery spending, infrastructure siting and insurance assessment — with the uncertainty stated.

Strategic

SDG 11 and SDG 13. Supports, and does not duplicate, Oman's national early-warning mandate. Methods use global open data and transfer to any Arab country with wadis and sabkhas.

One method family, three hazards already running: flood, drought and coastal change — each with its detection limit printed beside the result.

NEXT STEPS FOR INCUBATION

From a reproducible result to a deployed service

1

Independent validation

Measure one flood map against a third-party product and reported impact sites.

2

Hyperspectral

First EnMAP / Satellite 813 indicator selectable and runnable in the workspace.

3

GIQ

Publish the analysis service on the GIQ marketplace.

4

Pilot

One Omani public body, one past event it knows well, judged against its own records.

Repository: README, runnable notebook, pinned requirements, sample input and output — clone, install, run all cells, under one minute.